Project Design Phase-II

Solution Requirements Specification

Student Name: Pranay Patil	**Enrollment No:** EN25EL3010049	**Maximum Marks:** 2 Marks
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Project Title: FORMULA 1 Race Intelligence Dashboard Analysis Report (2010 – 2026)		**Domain:** Data Analytics with Tableau

Functional & Non-Functional Requirements Specification

Req ID	Category	Requirement Description	Success Metric / Validation
FR-01	Data Ingestion	Ingest and model 6,464 race records from Excel without data loss	Record count = 6,464 across 17 seasons (2010–2026)
FR-02	Benchmarking	Provide multi-line championship trajectory comparison by constructor	Visual clarity of Red Bull & Mercedes benchmarks vs mid-tier
FR-03	Tyre Telemetry	Calculate pace degradation and cliff thresholds for Soft, Medium, Hard compounds	Heatmap reflects circuit type downforce variations
FR-04	Leaderboard	Display cumulative driver points, wins, podiums, and poles with sorting	Instant dynamic top-10 driver rankings
FR-05	Compliance Audit	Classify DNF incidents into 6 mechanical and incident failure types	Engine, Collision, Gearbox, Brakes, Hydraulics accurately mapped
NFR-01	Performance	Dashboard interactions and global filter updates must execute under 2 seconds	Tableau extract (.hyper) query execution < 1.5s
NFR-02	Usability	Dark motorsport telemetry UI with high contrast and intuitive navigation	Adheres to WCAG AA color accessibility standards
NFR-03	Portability	Packaged as a standalone Tableau Packaged Workbook (.twbx) for distribution	Opens seamlessly on Tableau Desktop and Tableau Public